

CHARLOTTESVILLE 2 SSE, VA, USA

WMO: 746010

Lat: 37.9975N

Lon: 78.4656W

Elev: 359

StdP: 97.09

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 03759

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.4	(c) -8.1	(d) -21.2	(e) 0.6	(f) -9.4	(g) -18.7	(h) 0.7	(i) -6.7	(j) 5.6	(k) 9.3	(l) 5.1	(m) 8.6	(n) 1.6	(o) N/A	(p) 0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.8	(c) 31.7	(d) 23.0	(e) 30.4	(f) 22.5	(g) 29.2	(h) 22.0	(i) 24.1	(j) 29.4	(k) 23.5	(l) 28.5	(m) 22.9	(n) 27.6	(o) 1.5	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.6	18.1	27.1	21.9	17.3	26.0	21.3	16.7	25.4	74.7	29.6	72.0	28.6	69.4	27.8	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.5	(b) 3.9	(c) 3.5	(e) -13.0	(f) 33.5	(g) 2.8	(h) 1.6	(i) -15.0	(j) 34.7	(k) -16.7	(l) 35.6	(m) -18.3	(n) 36.5	(o) -20.3	(p) 37.7		
(a) 4.5	(b) 3.9	(c) 3.5	(e) -14.9	(f) 25.3	(g) 2.8	(h) 0.5	(i) -16.9	(j) 25.6	(k) -18.5	(l) 25.9	(m) -20.1	(n) 26.1	(o) -22.1	(p) 26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	1.6	3.9	7.7	13.6	18.4	22.5	24.4	23.4	20.4	14.9	8.3	4.6
	DBStd	9.12	5.62	6.02	5.89	5.19	4.20	2.96	2.68	2.56	3.65	4.69	4.83	4.98
	HDD10.0	870	267	187	117	22	1	0	0	0	0	11	88	178
	HDD18.3	2348	520	406	331	158	54	4	0	1	20	128	301	425
	CDD10.0	2218	5	15	46	131	262	374	447	416	313	162	37	12
	CDD18.3	655	0	0	2	16	57	129	189	158	83	21	1	0
	CDH23.3	4276	0	0	8	101	293	826	1453	1023	475	94	2	0
	CDH26.7	1083	0	0	0	12	42	194	446	257	114	18	0	0
Wind	WSAvg	1.7	2.0	2.0	2.0	2.0	1.6	1.4	1.3	1.3	1.3	1.7	1.8	1.9
Precipitation	PrecAvg	1207	84	82	100	82	114	102	131	110	118	103	92	83
	PrecMax	1894	246	204	210	249	268	325	377	266	456	323	332	188
	PrecMin	772	5	5	5	22	37	11	28	25	16	0	14	9
	PrecStd	237	48	48	46	46	51	72	75	53	98	75	63	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	21.1	24.5	28.4	29.6	32.3	34.0	32.5	31.7	28.8	22.7	19.7
		MCWB	12.7	14.7	14.4	19.1	20.6	23.6	24.0	22.2	22.0	21.0	16.0	15.5
	2%	DB	15.1	18.0	21.6	26.2	27.9	30.4	32.2	30.7	29.6	25.8	19.1	17.0
		MCWB	11.0	12.3	13.6	16.8	20.4	22.3	23.5	23.0	21.4	19.4	13.5	14.0
	5%	DB	12.8	15.7	19.4	24.1	26.5	28.9	30.6	29.4	27.9	23.7	17.4	14.8
		MCWB	8.6	10.7	12.1	16.2	20.0	21.8	22.9	22.4	21.0	18.5	11.9	11.4
	10%	DB	9.9	13.1	17.1	21.9	25.0	27.5	29.2	28.0	26.1	21.8	15.8	12.3
		MCWB	5.9	8.1	11.0	14.6	18.9	21.0	22.5	21.7	20.1	17.3	10.7	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.3	15.8	16.8	20.4	22.8	24.1	25.4	24.7	23.6	21.9	17.1	17.6
		MCDB	16.5	19.4	21.7	23.3	27.0	30.7	31.3	29.5	27.5	24.9	20.9	18.6
	2%	WB	12.2	14.2	15.2	18.3	21.7	23.1	24.6	23.7	22.7	21.0	15.6	15.0
		MCDB	14.4	17.3	19.5	23.5	26.3	28.6	30.1	28.3	27.2	24.5	18.9	16.6
	5%	WB	9.2	11.6	14.0	16.8	20.9	22.4	23.9	23.0	21.8	19.9	13.8	12.8
		MCDB	12.3	14.8	17.8	21.9	25.2	27.4	29.2	27.3	26.2	22.7	16.2	15.0
	10%	WB	7.1	9.3	12.3	15.5	20.1	21.7	23.2	22.4	20.9	17.9	11.8	10.1
		MCDB	10.2	12.9	16.5	20.1	24.2	26.1	28.2	26.3	24.2	21.2	14.7	12.2
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.9	9.5	9.7	8.2	7.9	7.8	7.7	7.6	7.7	7.9	7.2
		MCDBR	10.7	11.9	11.6	11.1	9.3	8.8	9.0	9.0	9.6	8.6	9.3	9.5
		MCWBR	8.6	8.2	6.8	5.8	5.2	4.9	4.0	4.0	4.0	4.4	5.8	7.3
	5% WB	MCDBR	10.1	10.6	10.6	10.4	8.5	8.3	8.1	7.8	7.8	7.4	8.8	8.5
		MCWBR	9.0	8.7	6.7	6.5	4.9	4.9	4.1	4.1	4.1	4.8	6.6	7.7
Clear-Sky Solar Irradiance	taub	0.290	0.306	0.339	0.378	0.420	0.466	0.487	0.492	0.394	0.344	0.310	0.293	
	taud	2.528	2.466	2.422	2.332	2.230	2.143	2.111	2.072	2.339	2.473	2.527	2.550	
	Ebn at Noon	904	927	924	900	864	821	800	785	853	871	869	875	
	Edh at Noon	76	92	107	124	140	153	156	159	115	91	76	70	
All-Sky Solar Radiation	RadAvg	2.26	3.05	4.05	5.04	5.61	6.17	5.87	5.29	4.38	3.43	2.53	1.98	
	RadStd	0.18	0.35	0.25	0.39	0.48	0.41	0.34	0.26	0.50	0.39	0.18	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+125	+72

Nomenclature: See separate page